

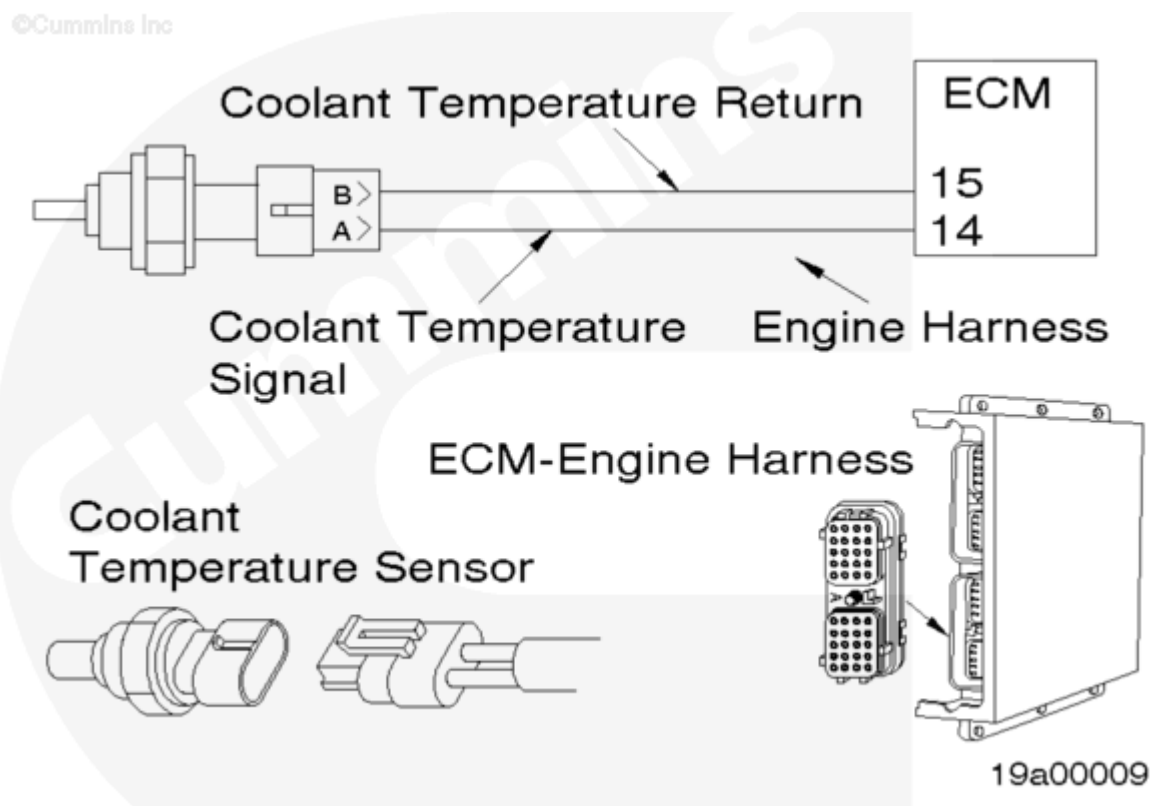
Fault Code: 144

Coolant Temperature Sensor (CTS) Circuit

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 144 PID(P): SPN: FMI: Lamp: SRT: 00-355	High voltage detected at engine coolant temperature sensor signal pin 14 of the engine harness Electronic Control Module (ECM) Connector.	None on performance. Common Warning output is energized.



Circuit Description

The CTS is used by the ECM to monitor the temperature of the engine coolant. The coolant temperature is used by the ECM for the engine protection system and fueling control. The ECM monitors the voltage on pin 14. The ECM expects to see the voltage vary between 0.32

and 4.69 VDC. If the voltage is above 4.60 VDC, then the ECM will log Fault Code 144. Voltage above 4.60 VDC on pin 14 can be caused by, opens in the signal or return wires, voltage shorts to the signal or return wires, or a failed open sensor.

Component Location

The CTS is located on the side of the thermostat housing.

Shoptalk

If the coolant temperature is below -18° C [0° F] the engine should be warmed and checked to see if fault goes inactive.

All Temperature Sensors

- The resistance of the sensor varies with the temperature. The reading that you observe will compare to the following table if the sensor is functioning properly.

NOTE: High voltages correspond to low temperatures and low voltages correspond to high temperatures.

Temperature	Temperature	Resistance
(° C)	(° F)	(ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Warnings and Cautions

WARNING

Wait until the coolant temperature is below 50° C [120° F] before removing the coolant system pressure cap or the CTS. Failure to do so can cause personal injury from heated coolant spray.

CAUTION

To avoid pin and harness damage, use the following test leads when taking a measurement:
Part No. 3822758 - male Deutsch/Cannon/Metri-Pack test lead
Part No. 3823256 - Metri-Pack 2-way connector test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the CTS.	
	STEP 1A. Inspect the harness and the sensor connector pins.	No damaged pins
	STEP 1B. Check the resistance of the CTS.	600 ohms to 36k ohms See temperature/resistance table under shop talk for correct value.
STEP 2.	Check the engine harness.	
	STEP 2A. Inspect the ECM and the engine harness adaptor cable connector pins.	No damaged pins
	STEP 2A-1. Inspect the engine harness and the engine harness extension cable(s) connector pins.	No damaged pins
	STEP 2B. Check for an open in the signal and return wires.	Less than 10 ohms
	STEP 2B-1. Check for an open in the engine harness adaptor cable and any engine harness extension cable used.	Less than 10 ohms
	STEP 2C. Check for a short circuit from the signal and return pins to all other pins.	More than 100k ohms
	STEP 2C-1. Check for a short circuit from pin to all other pins in the engine harness adaptor cable and any engine harness extension cable used.	More than 100k ohms
STEP 3.	Clear the fault code.	
	STEP 3A. Disable the fault code.	Fault Code 144 inactive
	STEP 3B. Clear the inactive fault code.	All faults cleared

STEP 1. Check the CTS.

STEP 1A. Inspect the harness and the sensor connector pins.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller **not** in the diagnostic mode.
- Disconnect the engine harness from the CTS.

Action	Specification/Repair	Next Step
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	No damaged pins	1B
	Repair the damaged pins Repair or replace the engine harness or the CTS, whichever has the damaged pins. • Repair the engine harness. Refer to Procedure 019-202 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the engine harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the CTS. Refer to Procedure 019-019 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Dry connector by using electrical contact cleaner, Part No. 3824510. • Replace the connector seal.	3A
• bent or broken pins • pushed back or expanded pins • corroded pins • moisture in or on the connector • missing connector seal.		

STEP 1B. Check resistance of CTS.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller **not** in the diagnostic mode.
- Disconnect the engine harness from the CTS.

Action	Specification/Repair	Next Step
• Measure the resistance between the two pins on the sensor side of the CTS connection.	600 ohms to 36k ohms	2A
	Replace the CTS Refer to Procedure 019-019 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.	3A

STEP 2. Check the engine harness.

STEP 2A. Inspect the ECM and the harness connector pins.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller **not** in the diagnostic mode.
- Disconnect the engine harness adaptor cable from the ECM.

Action	Specification/Repair	Next Step
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• bent or broken pins • pushed back or expanded pins • corroded pins • moisture in or on the connector.	No damaged pins	2A-1
	Repair the damaged pins Repair or replace the engine harness or the ECM, whichever has the damaged pins. • Repair the engine harness. Refer to Procedure 019-240 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the engine harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the ECM. Refer to OEM Procedures . • Dry the connector by using electrical contact cleaner, Part No. 3824510.	3A

STEP 2A. Inspect the ECM and the harness connector pins.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller **not** in the diagnostic mode.
- Disconnect the engine harness adaptor cable from the ECM.

Action	Specification/Repair	Next Step
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	No damaged pins	2A-1
	Repair the damaged pins Repair or replace the engine harness adaptor cable or the ECM, whichever has the damaged pins. • Repair the engine harness adaptor cable. Refer to Procedure 019-240 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the engine harness adaptor cable. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the ECM. Refer to OEM Procedures . • Dry the connector by using electrical contact cleaner, Part No. 3824510.	3A
• bent or broken pins • pushed back or expanded pins • corroded pins • moisture in or on the connector.		

STEP 2B. Check for an open in the signal and return wires.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller **not** in the diagnostic mode.
- Disconnect the engine harness adaptor cable from the ECM.
- Disconnect the engine harness from the CTS.

Action	Specification/Repair	Next Step
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• Measure the resistance from pin 14 of the engine harness adaptor cable connector to pin A on the harness side of the CTS connector.	Less than 10 ohms	2C
• Measure the resistance from pin 15 of the engine harness adaptor cable connector to pin B on the harness side of the CTS connector.		2B-1

STEP 2B-1. Check for an open in the engine harness adaptor cable and any engine harness extension cable used.

Conditions : • Stop/Run switch in the "STOP" position. • Controller not in the diagnostic mode. • Disconnect the engine harness adaptor cable from the ECM. • Disconnect the engine harness from the CTS.		
Action	Specification/Repair	Next Step

• Measure the continuity for pin 14 of the engine harness adaptor cable connector and any engine harness expansion cable used. • Measure the continuity for pin 15 of the engine harness adaptor cable connector and any engine harness expansion cable used.	Less than 10 ohms Repair or replace the engine harness • Repair the engine harness. Refer to Procedures 019-199, 019-202 and 019-240 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the engine harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.	3A
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Repair or replace the engine harness adaptor cable or engine harness extension cable(s), whichever is found faulty

- Repair the engine harness adaptor cable or engine harness extension cable(s). Refer to Procedures 019-199, 019-202 and 019-240 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.
- Replace the engine harness adaptor cable or engine harness extension cable(s). Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.

3A

STEP 2C. Check for a short circuit from the signal and return pins to all other pins.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller **not** in the diagnostic mode.
- Disconnect the engine harness adaptor cable from the ECM.
- Disconnect the engine harness from the CTS.

Action

Specification/Repair

Next Step

• measure the resistance from pin 14 in the engine harness adaptor cable connector to all other pins in the connector • measure the resistance from pin 15 in the engine harness adaptor cable connector to all other pins in the connector.	More than 100k ohms	3A
		2C-1

STEP 2C-1. Check for a short circuit from pin to all other pins in the engine harness adaptor cable and any engine harness extension cable used.

Conditions : • Stop/Run switch in the "STOP" position. • Controller not in the diagnostic mode. • Disconnect the engine harness adaptor cable from the ECM. • Disconnect the engine harness from the engine harness extension cable(s).		
Action	Specification/Repair	Next Step

• measure the resistance from pin 14 in the engine harness adaptor cable connector and any engine harness extension cables used, to all other pins in the connector • measure the resistance from pin 15 in the engine harness adaptor cable connector and any engine harness extension cables used, to all other pins in the connector	More than 100k ohms Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-199 and 019-240 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184. • Replace the engine harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.	3A
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Repair or replace the engine harness adaptor cable or engine harness extension cable(s), whichever is found faulty

- Repair the engine harness adaptor cable or engine harness extension cable(s). Refer to Procedures 019-199, 019-202 and 019-240 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.
- Replace the engine harness adaptor cable or engine harness extension cable(s). Refer to Procedure 019-043 in the Troubleshooting and Repair Manual QST Fuel System QST30 G-Drive Engine Series, Bulletin No. 3666184.

3A

STEP 3. Clear the fault code.

STEP 3A. Disable the fault code.

Conditions :

- Connect all components.

Action

Specification/Repair

Next Step

• Connect all components. • Start the engine and let idle for one minute. • Verify Fault Code 144 is inactive.	Fault Code 144 inactive	3B
	Return to troubleshooting steps or contact your local Cummins Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 3B. Clear the inactive fault codes.

Conditions :

- Connect all components.

Action	Specification/Repair	Next Step
• Erase the inactive fault codes using INSITE™, Part No. 3825145. NOTE: The datalink connector is located on the right bank of the flywheel housing.	All faults cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting chart